Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

Boris Garbuzov

Part 8 / 12

The Envelope Theorem and a Historical Note on Score vs LM

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1 The Envelope Theorem and Its Use in the LM Test

1.1 The envelope theorem in the simplest possible case

Before using the envelope theorem to generalise the shadow-price finding, we state and prove it in the most elementary setting: an *unconstrained* one-variable maximisation with a parameter.

Setup. Let $f(x, c)$ be a smooth function of two variables, x the variable being optimised and c a parameter. For each c, let

$$x^*(c) = \arg\max_x f(x, c), \qquad V(c) = f(x^*(c), c) = \max_x f(x, c)$$

be the maximiser and the resulting maximal value (the *value function*). Assume $x^*(c)$ is differentiable in c.

Question. How does $V(c)$ change as c changes? Naively, V depends on c in two ways: directly through the second argument of f, and indirectly because the maximiser $x^*(c)$ itself shifts. The envelope theorem says the second channel contributes nothing.

> **Envelope theorem (elementary form).**
>
> $$V'(c) = \frac{\partial f}{\partial c}\Big(x^*(c), c\Big).$$
>
> The total derivative of V equals the *partial* derivative of f with respect to c, evaluated at the optimum — with no correction term for dx^*/dc.

Proof. By the chain rule,

$$V'(c) = \frac{\partial f}{\partial x}\big(x^*(c), c\big) \cdot \frac{dx^*}{dc} + \frac{\partial f}{\partial c}\big(x^*(c), c\big).$$

Since $x^*(c)$ maximises $f(\cdot, c)$, the first-order condition gives $\partial f/\partial x\,(x^*(c), c) = 0$. The first term vanishes, leaving $V'(c) = \partial f/\partial c\,(x^*(c), c)$. $\square$

Worked example. Let $f(x, c) = -(x - c)^2 + c$. For fixed c, this is maximised at $x^*(c) = c$, giving $V(c) = f(c, c) = c$. Direct differentiation: $V'(c) = 1$.

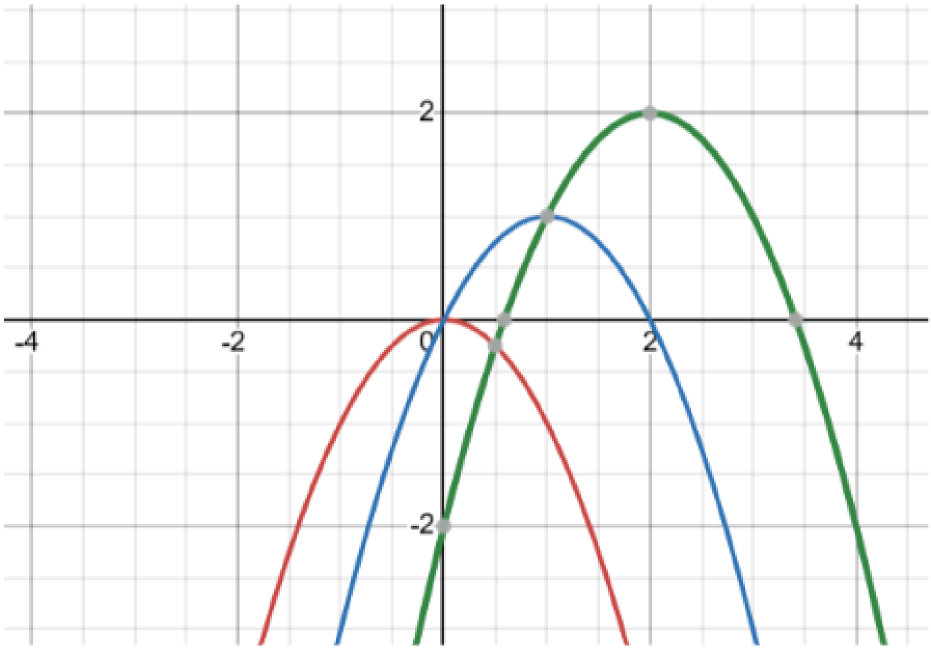

Figure 1: Three sections of $f(x, c) = -(x - c)^2 + c$ at $c = 0, 1, 2$ (red, blue, green): $-(x - 0)^2 + 0$, $-(x - 1)^2 + 1$, $-(x - 2)^2 + 2$. Each parabola peaks at $x^* = c$ with peak height $V(c) = c$: the peaks trace out the line $V(c) = c$ as c varies, visibly rising by one unit for each unit step in c.

Now check via the envelope theorem without re-deriving $x^*(c)$: $\partial f/\partial c = 2(x - c) + 1$. Evaluated at $x = x^*(c) = c$: $2(c - c) + 1 = 1$. Matches. The term $2(x - c)$, which would carry the effect of x^* shifting, evaluates to exactly zero at the optimum — this is the first-order condition doing its job.

On the definition of V and where it can fail. A precise statement of V is worth pausing on:

$$V(c) = \max_x f(x, c) = f(x^*(c), c).$$

If $x^*(c)$ is not defined at some $c = c_0$ (no maximiser exists, or it is not unique and we have not specified a selection rule), then $V(c_0)$ is simply not defined either, and the envelope-theorem formula cannot be applied there. The theorem is a statement about differentiating V *where V is well-defined and differentiable* — it presupposes, rather than establishes, the existence and smoothness of $x^*(\cdot)$ near c_0. In our example this is not an issue: $f(x, c) = -(x - c)^2 + c$ is a downward parabola in x for every c, so $x^*(c) = c$ exists and is unique and smooth everywhere.

The geometry in 3-d. The same example lets us see the envelope theorem geometrically.

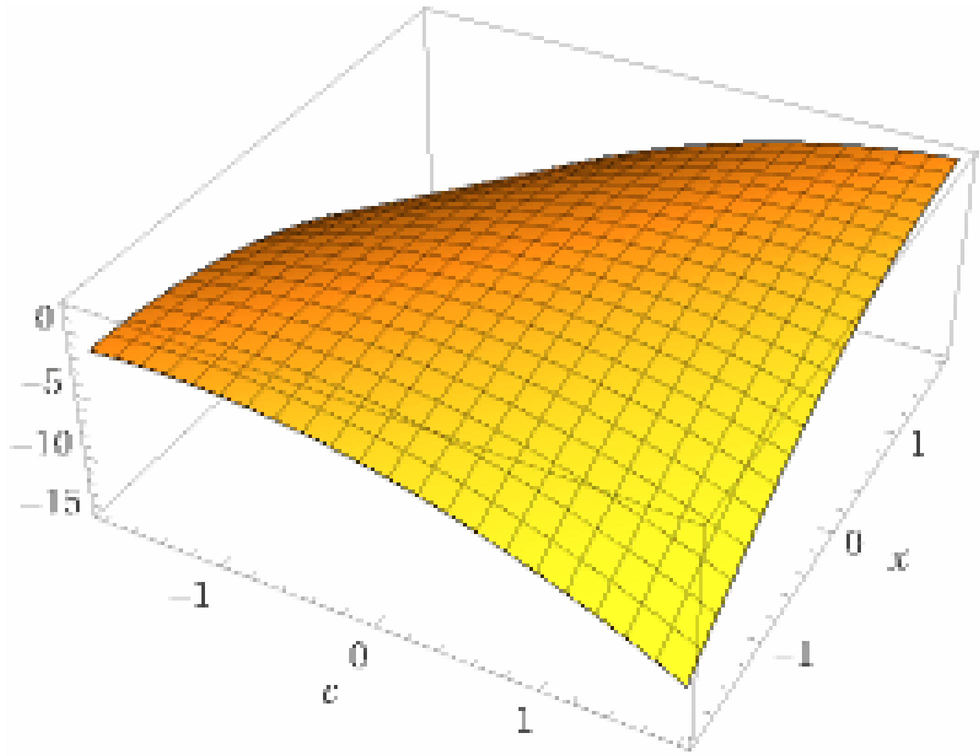

Figure 2: The surface $z = f(x, c) = -(x - c)^2 + c$ over the (x, c) plane. Every cross-section at fixed c (a slice parallel to the x-axis) is a downward parabola; its peak lies along the ridge $x = c$. The value function $V(c)$ is the height of this ridge, read off as the curve traced by the peaks as c varies — the "envelope" of the family of parabolic sections, which gives the theorem its name.

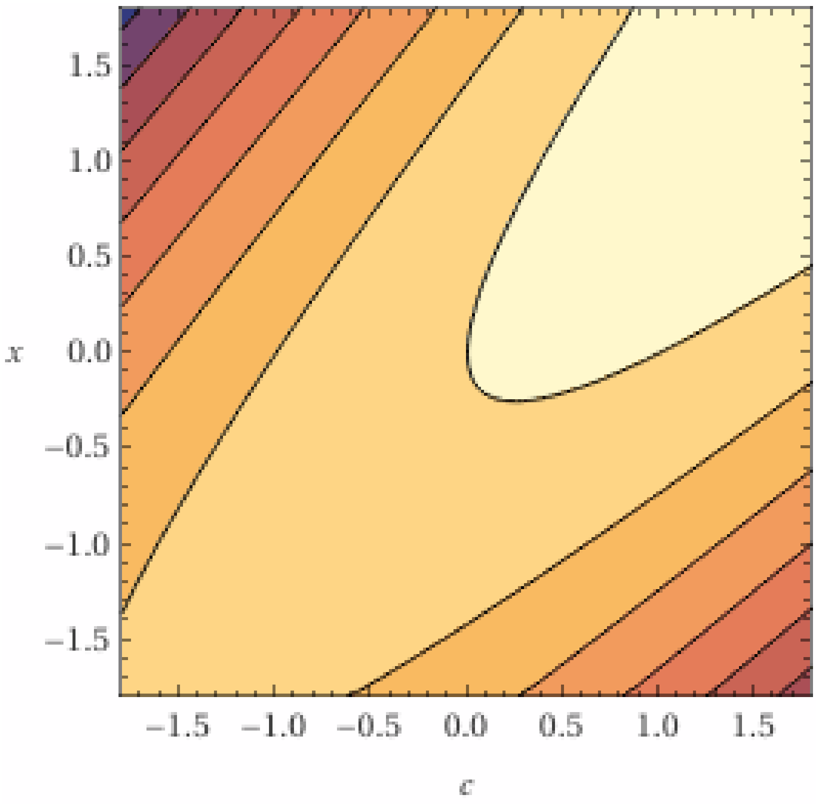

Figure 3: Contour (level-line) plot of $f(x,c) = -(x-c)^2 + c$ in the (c,x) plane. The ridge line $x = c$ (where each c-section attains its maximum) appears as the locus where the contours are tangent to vertical lines $c = \text{const}$ — i.e. where $\partial f/\partial x = 0$. Along this ridge, the contour values increase steadily with c, visibly confirming $V(c) = c$.

Interpretation. At the optimum, f is locally flat in the x-direction (that is what "optimum" means). So a small perturbation of c changes V exactly as if x^* had stayed put: the indirect effect through $x^*(c)$ is a second-order (negligible) correction, not a first-order one. This is the entire content of the theorem, and it generalises unchanged to the constrained case via the Lagrangian, as we use next.

1.2 An operator viewpoint, and a graphical proof of the equality

Two natural operators. Define two operators acting on functions $f(x,c)$ of two variables:

$$D_c : f \mapsto \frac{\partial f}{\partial c} \qquad \text{(partial differentiation in } c\text{)}, \qquad M_x : f \mapsto \max_x f(x,\cdot) \qquad \text{(maximise out } x\text{, leaving a func}$$

Applied to $f(x,c) = -(x-c)^2 + c$ in the two possible orders:

$$D_c\big(M_x f\big) = D_c V = V'(c) = 1, \qquad M_x\big(D_c f\big) = \max_x \big[\, 2(x-c) + 1 \,\big] = +\infty$$

(the second expression is unbounded in x, since $2(x-c)+1 \to +\infty$ as $x \to \infty$). So the two operators manifestly do *not* commute: $D_c M_x f \neq M_x D_c f$ in general — the right-hand expression need not even be finite.

What the envelope theorem actually says. The correct statement is not an order-of-operations identity between D_c and M_x applied independently, but a much more specific fact: D_c applied *after*

M_x equals $D_c f$ *evaluated at the particular point* $x = x^*(c)$ *singled out by* M_x — not maximised again, just evaluated there:

$$D_c\big(M_x f\big)(c) \;=\; \big(D_c f\big)\Big|_{x=x^*(c)},$$

i.e. $V'(c) = \partial f/\partial c\,(x^*(c), c)$. This is the boxed formula from the previous subsection, restated in operator language. It holds not because D_c and M_x commute as abstract operators, but because of the special structure at $x^*(c)$: the chain-rule term that would obstruct commutativity, $\partial f/\partial x \cdot dx^*/dc$, vanishes identically there by the first-order condition. "Commutativity" is the wrong frame; "evaluation at the optimiser, not re-optimisation" is the right one.

The graphical equality: tangency, not coincidence. The cleanest way to see $V'(c) = \partial f/\partial c\,(x^*(c), c)$ geometrically is to compare $V(c)$ against a *naive* slice of f at a fixed value of x — say $x = 1$ — treated as a function of c alone:

$$h(c) := f(1, c) = -(1-c)^2 + c.$$

h and V are *different* functions: $V(c) = c$ is the true value function (re-optimising x at every c), while $h(c)$ holds x fixed at 1 and only lets c move. They agree only where the fixed point $x = 1$ happens to equal the true optimiser, i.e. at $c = 1$ (since $x^*(1) = 1$). At every other c, $x = 1$ is not optimal, so $h(c) < V(c)$ strictly.

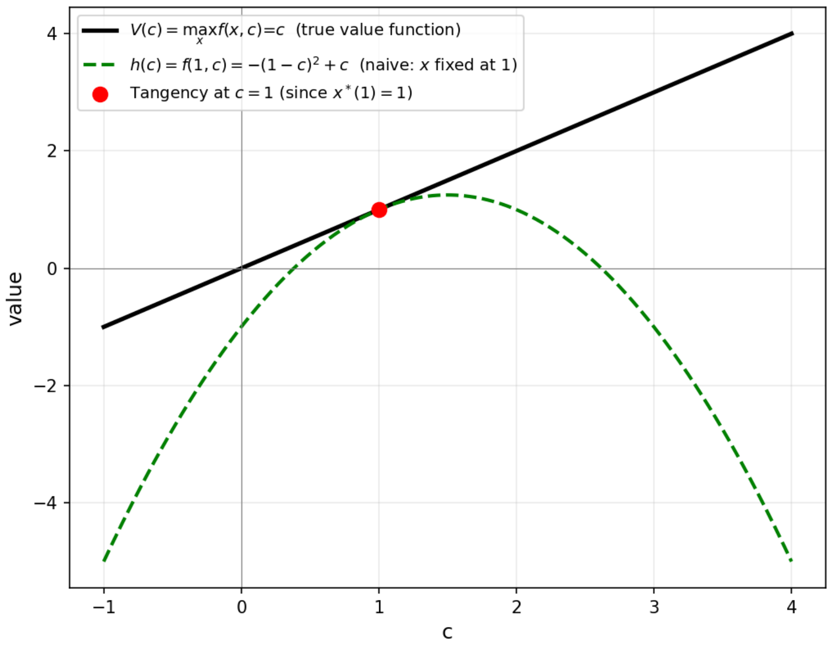

Figure 4: The true value function $V(c) = c$ (black, straight line) against the naive fixed-x slice $h(c) = f(1, c) = -(1-c)^2 + c$ (green, dashed parabola). The two curves touch only at $c = 1$ (red dot), where $x = 1$ coincides with the true optimiser $x^*(1) = 1$. Crucially, they are not just equal in value there but *tangent*: $V'(1) = h'(1) = 1$, since $h'(c) = 2(c-1) + 1$ gives $h'(1) = 1 = V'(1)$. Away from $c = 1$, the slopes differ. This is the geometric content of the envelope theorem: $\partial f/\partial c$ at the optimiser reproduces not just the value but the *slope* of V, at that one point only.

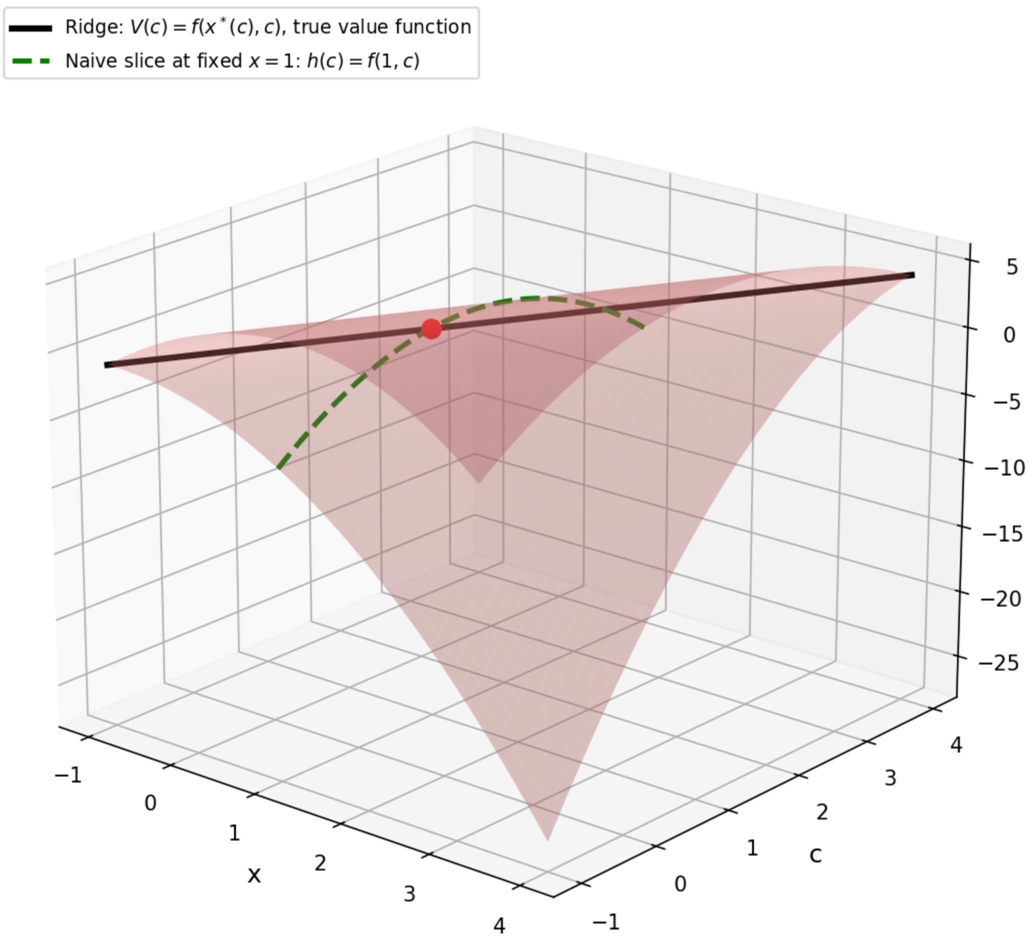

Figure 5: The same comparison lifted onto the surface $z = f(x, c) = -(x - c)^2 + c$. The black ridge is the true value function $V(c)$, traced by the vertex $x^*(c) = c$ of each cross-sectional parabola. The green dashed curve is the fixed-$x = 1$ slice $h(c) = f(1, c)$, lying *on* the surface but off the ridge except at one point. The red dot marks $(x, c) = (1, 1)$, where the green curve touches the black ridge — here $x = 1$ happens to be the optimiser for $c = 1$, so the two curves are tangent at this single point, with matching slope 1, even though they diverge everywhere else.

Why tangency, not equality of functions, is the right picture. This resolves a natural first guess: one might hope that $V(c)$ and some fixed-x slice $h(c) = f(x_0, c)$ are globally related (e.g. shifted versions of each other, as happens when two functions share *all* their partial derivatives). They are not — V and h are honestly different functions, equal and tangent at exactly one point (c such that $x^*(c) = x_0$) and diverging elsewhere. The envelope theorem is a statement about a single point of contact, not a global relationship between two surfaces or two curves: it identifies the slope of the ridge with the slope of *the one cross-sectional slice that happens to pass through the ridge at that point*, and no further.

A clean operator formula. The discussion above shows D_c and M_x do not commute as operators. But there *is* a clean operator identity hiding in the envelope theorem, once we introduce two further operators alongside D_c and M_x, with care about their domains and the types (signatures) of what goes in and what comes out.

- D_c: partial differentiation in the second slot.

$$D_c : f(x,c) \longmapsto g(x,c) := \frac{\partial f}{\partial c}(x,c).$$

  Type: (function of two variables) → (function of two variables). Natural domain: f differentiable in its second argument.

- M_x: maximise out the first slot.

$$M_x : f(x,c) \longmapsto h(c) := \max_x f(x,c).$$

  Type: (function of two variables) → (function of one variable). Natural domain: f bounded above in x for each c (so the maximum exists).

- A_x: the maximiser (argmax) in the first slot.

$$A_x : f(x,c) \longmapsto a(c) := \arg\max_x f(x,c).$$

  Type: (function of two variables) → (function of one variable). Natural domain: a strict subset of M_x's domain — we need not only a finite maximum but a *unique* maximiser, so that $a(c)$ is single-valued.

- S_x: substitution into the first slot.

$$S_x : \big(g(x,c),\, a(c)\big) \longmapsto k(c) := g\big(a(c), c\big).$$

  Type: (function of two variables, function of one variable) → (function of one variable). Domain: all pairs of the stated signatures (no further restriction); S_x is just composition, plugging $a(c)$ into the first slot of g and leaving c to appear in both places.

With these four operators precisely typed, the envelope theorem becomes the identity

$$\boxed{D_c\big(M_x f\big) \;=\; S_x\big(D_c f,\; A_x f\big).}$$

Why this is exactly right. Unwind both sides as functions of c:

$$\text{LHS:}\quad D_c(M_x f)(c) = \frac{d}{dc}\Big[\max_x f(x,c)\Big] = V'(c).$$

$$\text{RHS:}\quad S_x(D_c f,\, A_x f)(c) = (D_c f)\big(A_x f(c),\, c\big) = \frac{\partial f}{\partial c}\big(x^*(c),\, c\big).$$

These are literally the two sides of the boxed formula from §1.1: $V'(c) = \partial f/\partial c\,(x^*(c), c)$. So the formula is not a coincidental shuffling of symbols — it is a faithful, type-correct restatement of the envelope theorem, with $A_x f$ (not $M_x f$) feeding into the first slot of $D_c f$. Note carefully that f appears *twice* on the right: once differentiated ($D_c f$) and once maximised over via its argmax ($A_x f$) — these are two different transformations of the same underlying f, combined only at the very last step by S_x. This is the precise sense in which Question 1's intuition "it should be some kind of substitution into the derivative" is correct, once A_x (not the cruder M_x) is used to supply the point of substitution.

Remark 1. The earlier (incorrect) guess $M_x(D_c f)$ fails for two reasons simultaneously: (i) it maximises $D_c f$ over x at each c *independently* of where f itself is maximised, so it re-optimises rather than substitutes; and (ii) as the example showed, this re-optimisation can diverge to $+\infty$ even when $M_x f$ itself is perfectly well-defined. The corrected formula $S_x(D_c f, A_x f)$ avoids both problems: $A_x f$ locates the same point $x^*(c)$ used to build V, and S_x merely evaluates $D_c f$ there — no second optimisation is performed.

We showed $\lambda^* = f'(\theta_0)$, i.e. the multiplier is the derivative of the constrained optimum with respect to the constraint level. How far does this identification extend?

Setup: optimum as a function of the constraint level. Consider the family of problems indexed by c:

$$(\mathrm{P}_c): \qquad \max_{\theta} f(\theta) \quad \text{subject to} \quad g(\theta) = c.$$

For each c in some neighbourhood, let $\theta^*(c)$ denote the solution and define the *value function*

$$V(c) = f(\theta^*(c)) = \max_{\theta:\, g(\theta)=c} f(\theta).$$

The shadow price is $V'(c)$ — the rate at which the optimal value changes as the constraint level shifts. In economics this is the general object of interest: differentiate the optimised outcome with respect to any parameter of the problem, not only with respect to one written as a multiplier.

The Probabilist's derivation: $\theta^*(c)$ as a local inverse. Before invoking the envelope theorem directly, it is worth seeing *why* $\theta^*(c)$ is differentiable at all, following the Probabilist's argument explicitly. At the unconstrained optimum (no c yet), the first-order condition is

$$f'(\theta^*) - \lambda^* g'(\theta^*) = 0,$$

which, together with the constraint $g(\theta^*) = c$, can be solved implicitly for (θ^*, λ^*) as functions of c. The Probabilist's notes record this via the implicit function theorem applied directly to the stationarity system: writing $\theta^* = \theta^*(c)$ and differentiating $g(\theta^*(c)) = c$ with respect to c,

$$g'(\theta^*) \cdot \frac{d\theta^*}{dc} = 1 \implies \frac{d\theta^*}{dc} = \frac{1}{g'(\theta^*)}.$$

Equivalently: locally near the current c, $\theta^* = g^{-1}(c)$ — the constrained optimiser is (locally) the inverse function of the constraint map itself, since g is a bijection near a point where $g' \neq 0$. This is exactly the relation $\lambda^* = f'(\theta^*)/g'(\theta^*)$ combined with $\theta^* = g^{-1}(c)$ that appears in the handwritten notes.

Having established that $\theta^*(c)$ (and likewise $\lambda^*(c)$) are well-defined differentiable functions of c near the current value, we can now form the composite

$$y(c) := f\big(\theta^*(c)\big),$$

and ask for $y'(c)$. By the chain rule,

$$y'(c) = f'\big(\theta^*(c)\big) \cdot \frac{d\theta^*}{dc}(c).$$

This is precisely $V(c)$ from the paragraph above ($y \equiv V$), and substituting $d\theta^*/dc = 1/g'(\theta^*)$ recovers

$$V'(c) = \frac{f'(\theta^*(c))}{g'(\theta^*(c))} = \lambda^*(c),$$

matching the boxed formula for λ^* found earlier by direct substitution into the stationarity conditions. So the Probabilist's route (differentiate the implicit system directly) and the envelope-theorem route (below) arrive at the same conclusion; the envelope theorem simply packages the cancellation that happens inside this chain-rule computation, without requiring us to solve for $d\theta^*/dc$ explicitly.

Recovering λ via the envelope theorem. The elementary envelope theorem above had a single unconstrained variable x. To apply it to the Lagrangian, we enlarge the variable solved for via the first-order condition from θ alone to the pair (θ, λ), and treat the joint stationarity of $\mathcal{L}$ in this pair as the elementary theorem's stationary point x^* — *not* as a maximisation: as the determinant computation in Part 5b-a, §2.9 shows, $\mathcal{L}$ has a genuine saddle in (θ, λ), and for fixed $\theta \neq c$ it is in fact *unbounded* in λ (linear, with no extremum at all). So there is no optimisation over λ in any ordinary sense — the λ first-order condition $\partial\mathcal{L}/\partial\lambda = 0$ exists only to *recover the constraint*, not to optimise anything. This is legitimate because at a stationary point of a Lagrangian, *both* partial derivatives vanish — exactly the single first-order condition $\partial f/\partial x = 0$ used in the proof above, just written out in two components instead of one.

Form the Lagrangian for problem (P_c):

$$\mathcal{L}(\theta, \lambda; c) = f(\theta) - \lambda\,(g(\theta) - c).$$

At the solution, $\theta^*(c)$ and $\lambda^*(c)$ are both functions of c (assumed differentiable, at least locally near the current c). Differentiating $V(c) = \mathcal{L}(\theta^*(c), \lambda^*(c); c)$ with respect to c and using the envelope theorem (the first-order conditions $\partial_\theta\mathcal{L} = 0$ and $\partial_\lambda\mathcal{L} = 0$ kill the terms coming from $d\theta^*/dc$ and $d\lambda^*/dc$, exactly as the single term $\partial f/\partial x \cdot dx^*/dc$ vanished in the elementary case):

$$V'(c) = \left.\frac{\partial\mathcal{L}}{\partial c}\right|_{\theta^*(c),\,\lambda^*(c)} = \lambda^*(c).$$

So λ^* *is* $V'(c)$: the Lagrange multiplier is the derivative of the optimum with respect to the right-hand side of the constraint. This recovers our earlier finding $\lambda^* = f'(\theta_0)$ as the special case $g(\theta) = \theta$, $c = \theta_0$, where $V(c) = f(c)$ trivially and $V'(c) = f'(c)$.

How general is this? The derivation above did not use any special form of g — it holds for any smooth equality constraint $g(\theta) = c$, scalar or vector θ, as long as the envelope theorem's regularity conditions hold (differentiability of $\theta^*(\cdot)$, $\lambda^*(\cdot)$ near c, and the constraint qualification). So within the class of problems written as "maximise f subject to $g(\theta) = c$," the shadow price of relaxing c is always exactly λ^* — one of the Lagrange multipliers (the one attached to that constraint, if there are several).

Beyond constraint levels: shadow prices in general. In economics, "shadow price" is used more broadly: one differentiates the optimised value with respect to *any* parameter of the problem, not only a constraint level appearing on the right-hand side. For example, if f itself depends on a parameter a (say $f(\theta; a)$, as in a revenue function with a price parameter a), then $\partial V/\partial a$ is also called a shadow price, but it need not equal any Lagrange multiplier — the envelope theorem gives $\partial V/\partial a = \partial f/\partial a$ evaluated at θ^*, with no λ involved, because a does not appear in the constraint at all.

A further generalisation: parametrising g itself. The Probabilist's suggestion: under suitable regularity, one can go further and let the constraint be written as $g(\theta, c) = 0$ directly — i.e. c enters the left-hand side, possibly nonlinearly, rather than appearing as a separate right-hand-side level subtracted off at the end. Write the family of problems as

$$(\mathrm{P}_c): \qquad \max_\theta f(\theta) \quad \text{subject to} \quad g(\theta, c) = 0.$$

This is strictly more general than (P_c) above: it reduces to it when $g(\theta, c) := g(\theta) - c$, but allows c to enter nonlinearly, or even to enter both θ and c jointly in ways that do not separate.

Assume, for each c, a unique maximiser $\theta^*(c)$ exists and is differentiable in c (if several local maxima exist, the same construction applies separately to each differentiable branch $\theta_i^*(c)$; we work with one branch and assume uniqueness for simplicity). Define the value function $f^*(c) := f(\theta^*(c))$, which is differentiable as a composition of differentiable functions. We re-derive the shadow-price formula by the Probabilist's direct route (differentiating the constraint), and then recover it again via the envelope theorem, to confirm the two routes agree.

Remark 2 (Branches and the manifold picture). The case of several local maxima, handled above by picking a differentiable branch $\theta_i^*(c)$, is the same phenomenon as Question 2's restricted function $\tilde{f}$ on the constraint manifold $\{g(\theta, c) = 0\}$ for fixed c: each branch $\theta_i^*(c)$ traces a local maximum of $\tilde{f}$ on that manifold, and as c varies these local maxima move continuously along it. Making this correspondence precise — describing the branches $\theta_i^*(c)$ intrinsically as critical points of $\tilde{f}$ varying with c, rather than extrinsically via the first-order condition in the ambient space — requires the implicit function theorem applied jointly in (θ, c), which is a genuinely more delicate tool than anything used here. We do not pursue this; for our purposes the extrinsic description (the first-order condition $\nabla_\theta \mathcal{L} = 0$ in the full θ-space, for each fixed c) is sufficient and is what the derivation below uses throughout.

Step 1: differentiate the constraint. Since $g(\theta^*(c), c) \equiv 0$ for every c, differentiate both sides with respect to c using the chain rule:

$$\frac{\partial g}{\partial \theta}\Big(\theta^*(c), c\Big) \cdot \frac{d\theta^*}{dc} + \frac{\partial g}{\partial c}\Big(\theta^*(c), c\Big) = 0.$$

Solve for $d\theta^*/dc$ (valid whenever $\partial g/\partial\theta \neq 0$, the constraint qualification):

$$\frac{d\theta^*}{dc} = -\left.\frac{\partial g/\partial c}{\partial g/\partial \theta}\right|_{(\theta^*(c),\, c)}.$$

Step 2: differentiate the value function. By the chain rule, $f^{*\prime}(c) = f'(\theta^*(c)) \cdot d\theta^*/dc$. Substituting the result of Step 1:

$$f^{*\prime}(c) = -\, f'(\theta^*(c)) \cdot \left.\frac{\partial g/\partial c}{\partial g/\partial \theta}\right|_{(\theta^*(c), c)}.$$

Step 3: bring in λ^*. From the Lagrangian first-order condition $f'(\theta^*) - \lambda^* \, \partial g/\partial\theta\,(\theta^*, c) = 0$, we have $f'(\theta^*) = \lambda^* \cdot \partial g/\partial\theta\,(\theta^*, c)$. Substituting into the Step 2 formula, the $\partial g/\partial\theta$ factors cancel:

$$\boxed{f^{*\prime}(c) = V'(c) = -\,\lambda^*(c)\,\frac{\partial g}{\partial c}\Big(\theta^*(c), c\Big).}$$

Before this cancellation we had a ratio of two partial derivatives of g; after substituting the Lagrangian condition we are left with just the numerator term, multiplied by λ^*.

Recovering the simpler case. To confirm this generalises the earlier finding, set $g(\theta, c) = \theta - c$ (our original convention). Then $\partial g/\partial c = -1$, and the boxed formula gives $V'(c) = -\lambda^* \cdot (-1) = \lambda^*$ — exactly the earlier result $\lambda^* = f'(\theta_0)$. So the simple case is the special instance where c enters g additively with coefficient -1.

Worked example: the multiplier is not parametrisation-invariant. Take $f(\theta) = -(\theta - 3)^2$ (unconstrained maximum at $\theta = 3$) and compare two constraints with the *same* zero set $\{\theta = c\}$ (for $\theta, c > 0$), written two different ways:

$$g_1(\theta, c) = \theta - c, \qquad g_2(\theta, c) = \theta^2 - c^2.$$

Both vanish exactly when $\theta = c$, so both give the same $\theta^*(c) = c$ and the same value function $V(c) = f(c) = -(c-3)^2$, hence the same true shadow price $V'(c) = 6 - 2c$ (computed directly, with no reference to either Lagrangian).

But the two multipliers differ:

$$\lambda_1^*(c) = \frac{f'(\theta^*)}{\partial g_1/\partial\theta} = \frac{6-2c}{1} = 6-2c, \qquad \lambda_2^*(c) = \frac{f'(\theta^*)}{\partial g_2/\partial\theta} = \frac{6-2c}{2c} = \frac{3-c}{c}.$$

Checking the boxed formula for each: for g_1, $\partial g_1/\partial c = -1$, so $V'(c) = -\lambda_1^* \cdot (-1) = 6-2c$ — correct. For g_2, $\partial g_2/\partial c = -2c$, so $V'(c) = -\lambda_2^* \cdot (-2c) = \frac{3-c}{c} \cdot 2c = 6-2c$ — also correct, by a different route. Both recover the same true shadow price, but $\lambda_1^* \neq \lambda_2^*$ for $c \neq 0, 3$: the Lagrange multiplier itself depends on how the constraint is *written*, even though the constraint set and the shadow price do not. Only $V'(c)$ is an invariant of the problem; λ^* is an artefact of the chosen parametrisation of g.

Connection to the Wikipedia formulation and to inequality constraints. The Wikipedia article on the envelope theorem treats this in slightly more general terms still: it allows f itself to depend on c (write $f(\theta, c)$, not just $f(\theta)$), and states the result for inequality-constrained problems, which requires Karush–Kuhn–Tucker machinery beyond what we need here. Stripped down to our equality-constraint, $f(\theta)$-independent-of-c case, and writing $V(c)$ in place of Wikipedia's value-function notation (matching our $f^*(c)$), their general statement specialises to exactly the boxed formula above. The cleanest way to state it, avoiding the need to re-derive Steps 1–3 each time: extend the Lagrangian itself to carry the c-dependence,

$$\mathcal{L}(\theta, \lambda; c) = f(\theta) - \lambda\, g(\theta, c),$$

and apply the elementary envelope theorem of §1.1 directly to this $\mathcal{L}$, treating (θ, λ) jointly as the stationary point "x^*" of the elementary theorem (not as a maximiser):

$$V'(c) = \left.\frac{\partial \mathcal{L}}{\partial c}\right|_{\theta^*(c), \lambda^*(c)} = -\lambda^*(c)\, \frac{\partial g}{\partial c}(\theta^*(c), c),$$

identical to the boxed result, obtained with no need to separately differentiate the constraint. This is the version that should be used in practice: state the parametrised Lagrangian once, then differentiate it in c at the optimum — the envelope theorem guarantees the $d\theta^*/dc$ and $d\lambda^*/dc$ terms vanish automatically, exactly as they did in the elementary case.

1.3 Building the chain rule graphically: the Probabilist's step-by-step picture series

The envelope-theorem discussion above used the chain rule in passing (§1.1), but only for a single variable x. In the Lagrangian setting, and in general, we need the chain rule for a function of *two* variables $f(x, y)$ composed with a parametric path $x = x(c)$, $y = y(c)$. Before specialising to the envelope theorem (where one of the two variables will turn out to be the identity, $y(c) = c$), the Probabilist's notes build up the general picture in small steps, each one illustrated separately. We reproduce the series here exactly, with commentary, and then continue it to the dot-product formula it is building toward.

Setup. We have a surface $z = f(x, y)$ (Figure 6) and a path in the (x, y)-plane given parametrically by $c \mapsto (x(c), y(c))$ (Figure 7). Importantly, y is not assumed to equal c here — both $x(c)$ and $y(c)$ are, for now, general functions of the parameter c. This generality matters: in the envelope-theorem application later, $y(c) = c$ will be a special (degenerate) case, and seeing the general chain rule first prevents confusing the special structure of that case with the general mechanism.

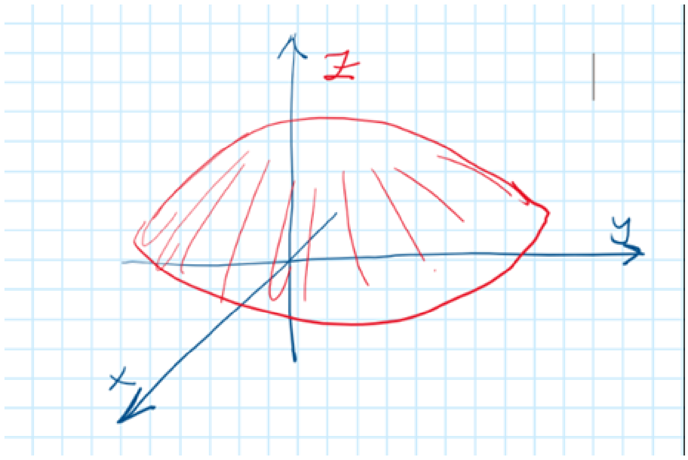

Figure 6: The surface $z = f(x, y)$. A generic dome shape, no parametrisation yet.

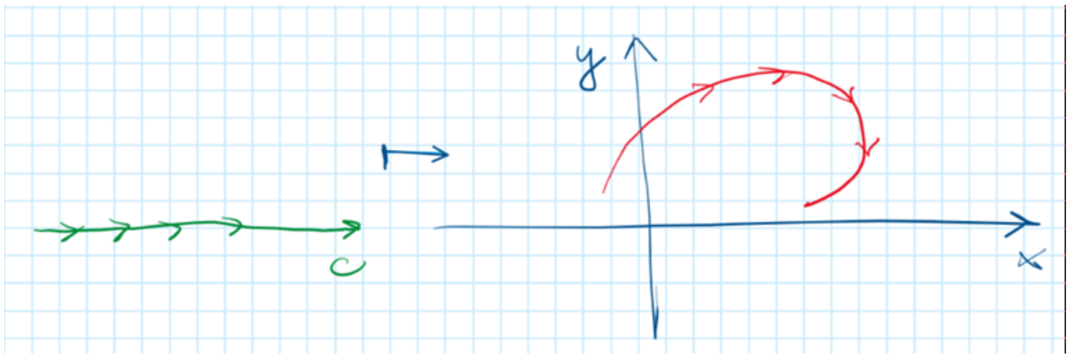

Figure 7: Left: the parameter axis c (green), with the direction of increasing c marked. Right: the image of the map $c \mapsto (x(c), y(c))$ in the (x, y)-plane (red curve, arrows showing the direction of increasing c along the path).

The path as a curve in (c, x, y)-space, and its three shadows. To keep track of both component functions $x(c)$ and $y(c)$ simultaneously, the Probabilist lifts the picture into the three-dimensional space with axes (c, x, y) — note this is *not* the (x, y, z) space of the surface f; $z = f$ has not entered yet. The red space curve is $c \mapsto (c,\ x(c),\ y(c))$, and it casts three shadows (orthogonal projections) onto the three coordinate planes.

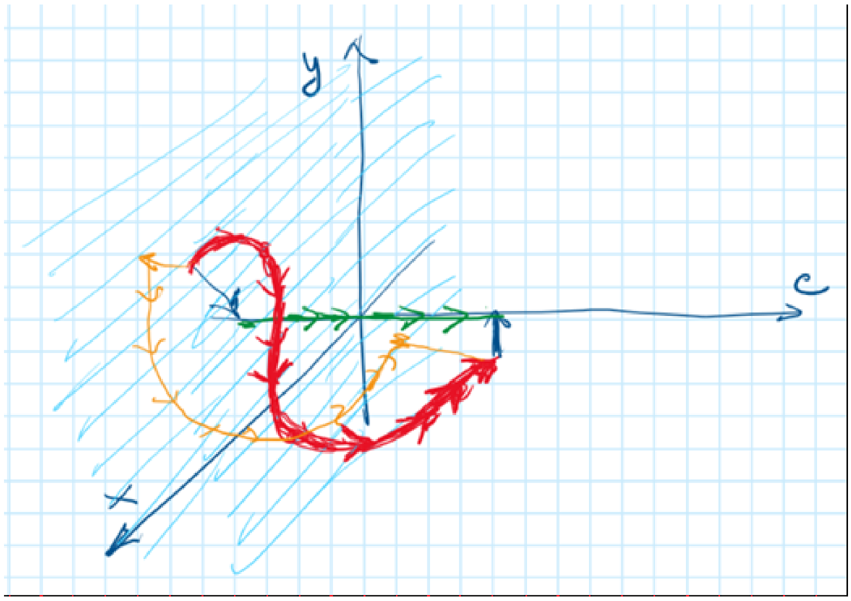

Figure 8: The space curve $(c, x(c), y(c))$ in red, together with its shadow (orange) on the (x, y)-plane — this orange shadow is exactly the red curve of Figure 7, now seen as a projection rather than drawn directly.

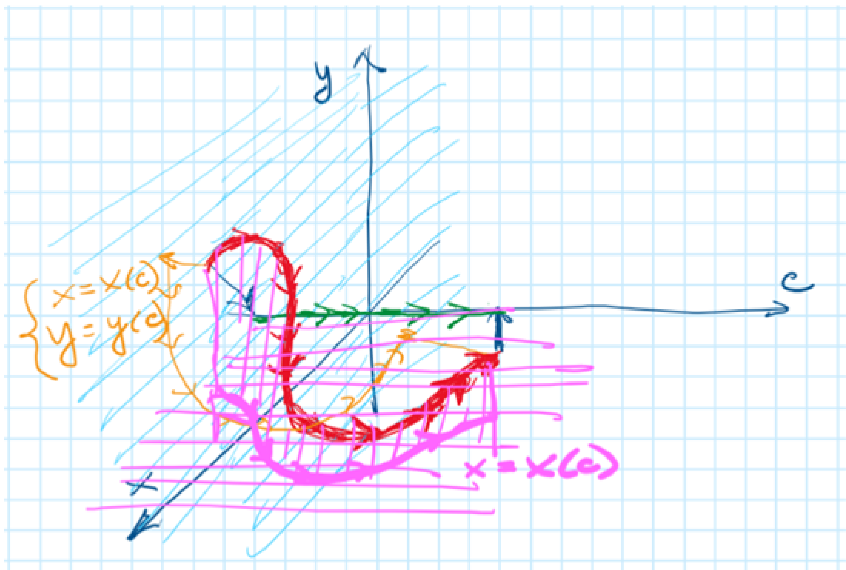

Figure 9: A second shadow added: the magenta projection onto the (c, x)-plane is the graph of $x = x(c)$ — the first component function on its own, as an ordinary function of c.

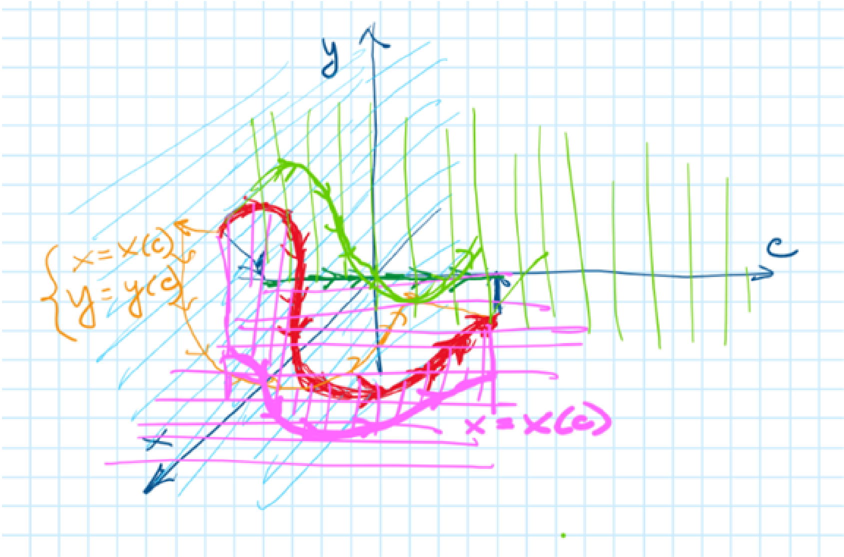

Figure 10: The third shadow added: the green projection onto the (c, y)-plane is the graph of $y = y(c)$. All three shadows are now visible together: (x, y) (orange, the path itself), (c, x) (magenta, the graph of $x(\cdot)$), (c, y) (green, the graph of $y(\cdot)$). This completes the description of the parametric path and its two component functions.

The same construction for the derivatives. The Probabilist repeats the identical construction one level up: instead of the curve $(c, x(c), y(c))$, take the curve of *derivatives* $(c,\, x'(c),\, y'(c))$, with primed axes x', y'. The logic and the three shadows are exactly analogous — only the object being plotted has changed, from positions to velocities.

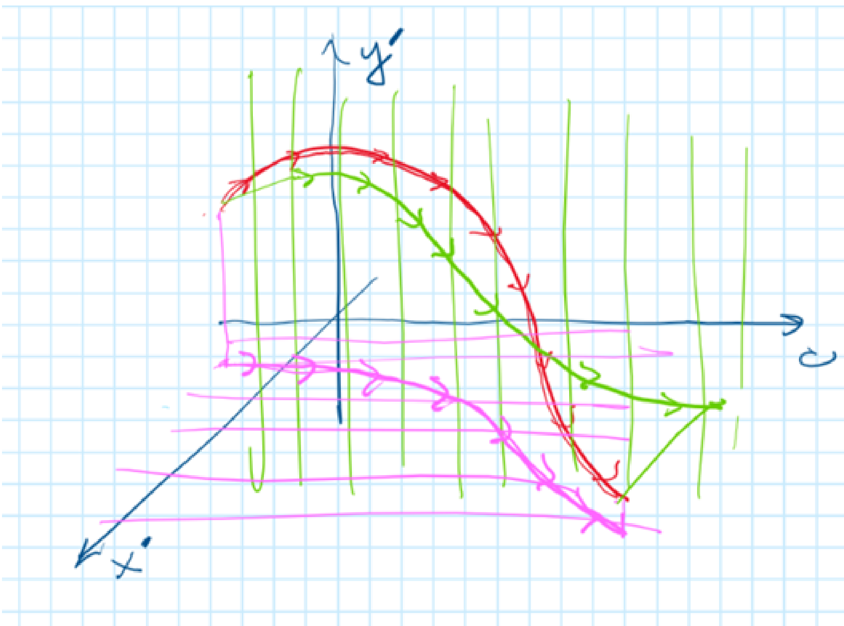

Figure 11: The same three-shadow construction, now for the derivative curve $(c, x'(c), y'(c))$: green is the (c, y')-shadow (graph of $y'(\cdot)$), magenta is the (c, x')-shadow (graph of $x'(\cdot)$).

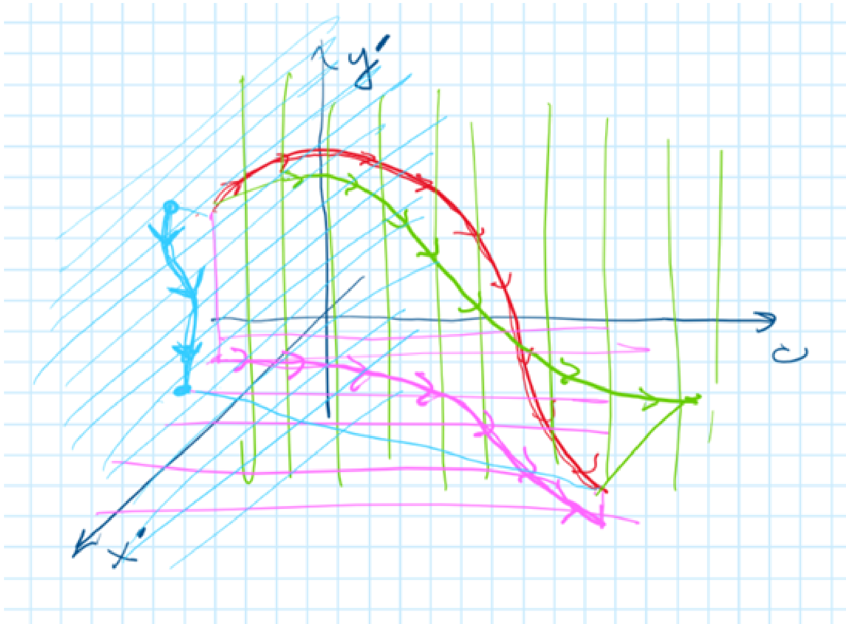

Figure 12: The fourth projection added, in light blue: the shadow onto the (x', y')-plane itself. This is the one we did not draw in the first (position) series — it is the new, essential object: the image of the velocity vector $(x'(c), y'(c))$, traced out as c varies.

Isolating the velocity vector. The (x', y')-shadow is now extracted on its own and reinterpreted: rather than viewing it as a curve traced by a moving point, each point $(x'(c_0), y'(c_0))$ on it is drawn as a vector emanating from the origin — the instantaneous *velocity* of the path $(x(c), y(c))$ at parameter value c_0.

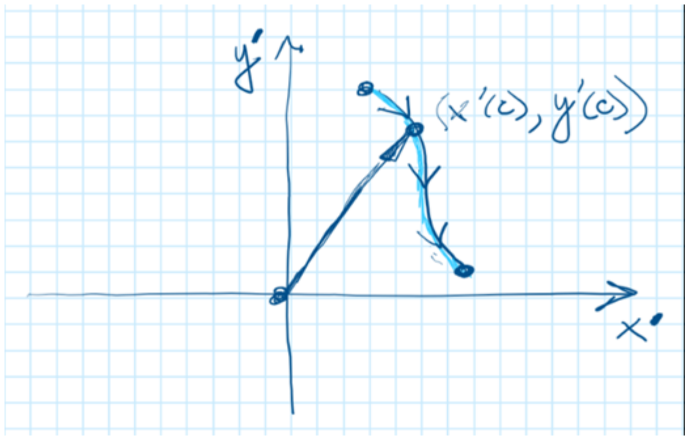

Figure 13: The point $(x'(c), y'(c))$ on the (x', y')-plane, shown as a radius vector from the origin. This is the velocity vector of the path at the corresponding parameter value — the object that will be dotted with the gradient of f below.

Returning to the surface: lifting the path onto f. The second half of the series returns to the surface $z = f(x, y)$ and the floor path $(x(c), y(c))$, now shown together and then combined: the floor path is lifted vertically onto the surface, producing the composite curve $c \mapsto \big(x(c),\, y(c),\, f(x(c), y(c))\big)$.

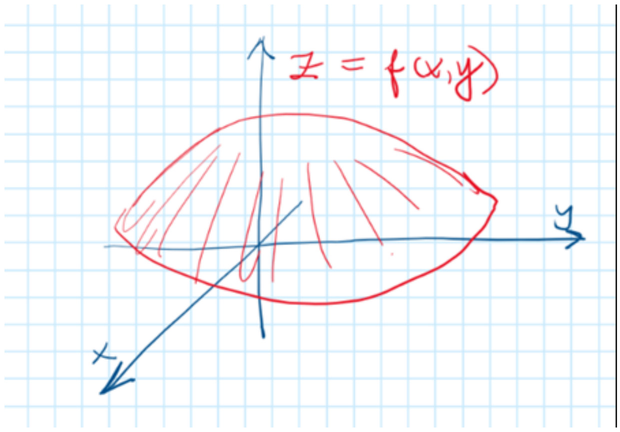

Figure 14: The surface $z = f(x, y)$, redrawn and explicitly labelled.

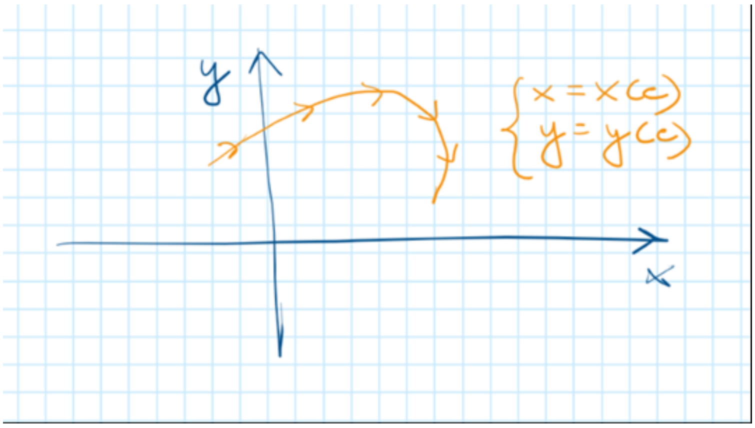

Figure 15: The floor path $\{x = x(c),\, y = y(c)\}$, redrawn in isolation, matching Figure 7.

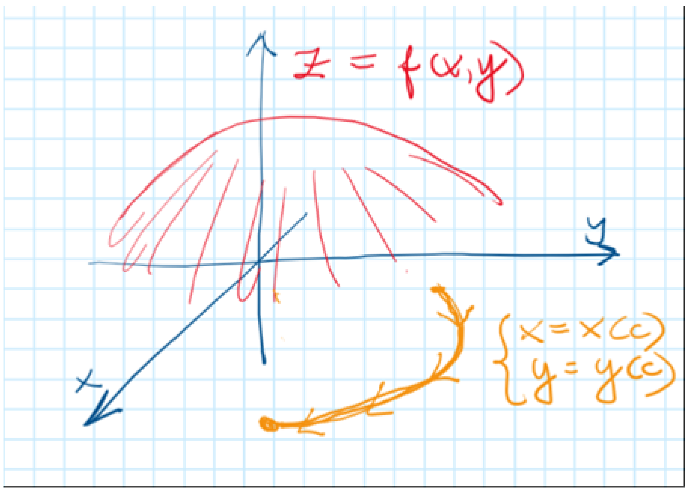

Figure 16: Surface and floor path shown together, the path still lying flat on the $z = 0$ plane beneath the surface — not yet lifted onto it.

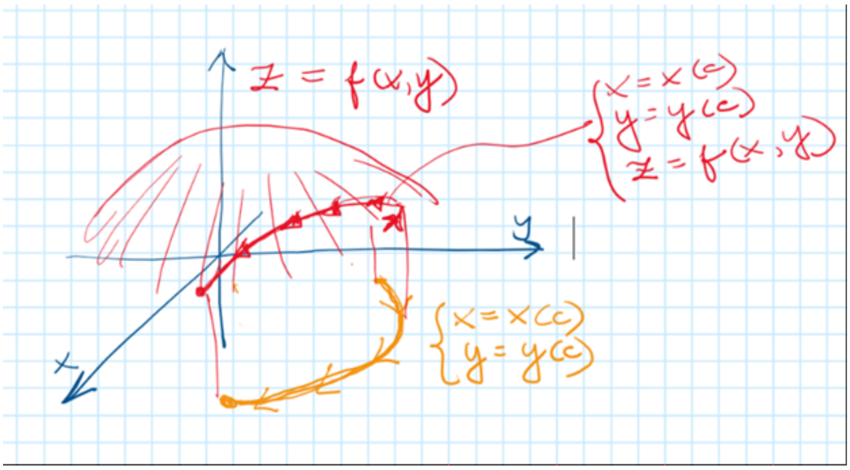

Figure 17: The floor path lifted vertically onto the surface, producing the red space curve $\{x = x(c),\, y = y(c),\, z = f(x, y)\}$ — i.e. the graph of $h(c) := f(x(c), y(c))$ traced out on the surface itself. This is the final object of the Probabilist's series: the composite function whose derivative we want, displayed not as an abstract formula but as an actual curve riding on the surface, directly above its shadow on the floor.

Continuing the series: assembling the chain rule. With every piece now in view — the surface, the floor path, the lifted curve $h(c)$, and the velocity vector $(x'(c), y'(c))$ — we can assemble them into the chain-rule formula. We continue the Probabilist's series with three more pictures, completing the construction he sketched and naming as the goal: *this is a dot product of two vectors.*

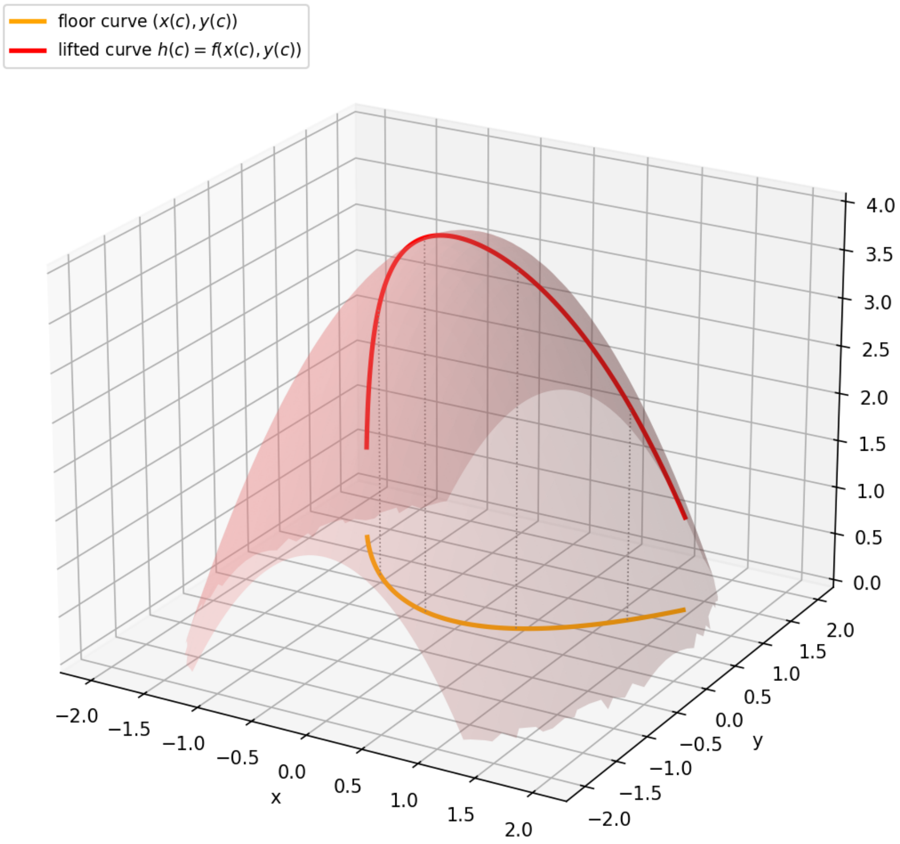

Figure 18: All the pieces together for a concrete example $f(x, y) = 4 - x^2 - 0.6y^2$ with path $x(c) = c$, $y(c) = \frac{1}{2}c^2$ (chosen so that y is *not* the identity, matching the Probabilist's emphasis that y need not equal c): the orange floor curve $(x(c), y(c))$, and the red lifted curve $h(c) = f(x(c), y(c))$ riding on the surface directly above it. Dotted vertical segments show the lift at a few sample parameter values. This is the direct continuation of Figure 17 to a fully worked numerical case.

The chain rule itself. $h(c) = f(x(c), y(c))$ is a composition of $f : \mathbb{R}^2 \to \mathbb{R}$ with the path $c \mapsto (x(c), y(c)) : \mathbb{R} \to \mathbb{R}^2$. The multivariate chain rule gives

$$h'(c) = \frac{\partial f}{\partial x}\big(x(c), y(c)\big) \cdot x'(c) \; + \; \frac{\partial f}{\partial y}\big(x(c), y(c)\big) \cdot y'(c).$$

Writing the two partial derivatives as the gradient vector $\nabla f = (\partial f/\partial x,\ \partial f/\partial y)$, evaluated at $(x(c), y(c))$, and the two derivatives $x'(c), y'(c)$ as the velocity vector from Figure 13, the right-hand side is exactly a dot product:

$$\boxed{h'(c) = \nabla f\big(x(c), y(c)\big) \cdot \big(x'(c),\, y'(c)\big).}$$

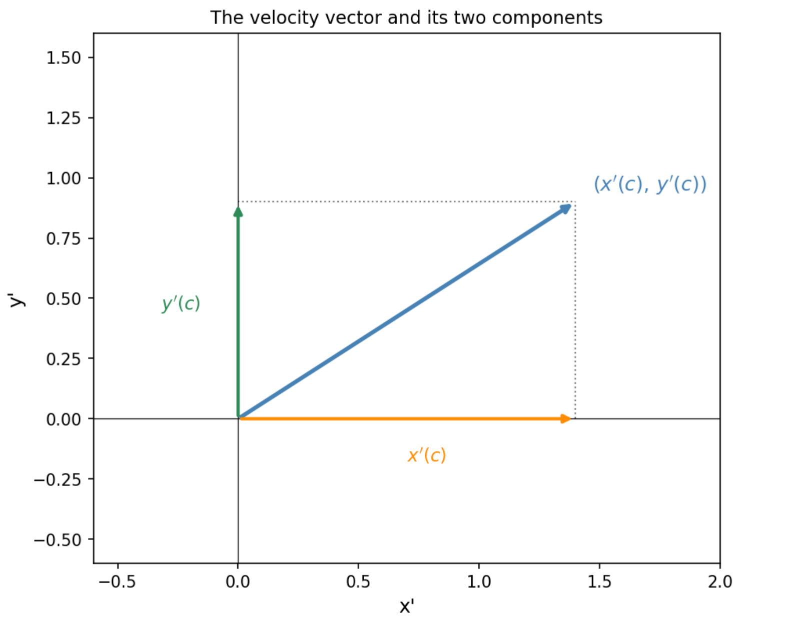

Figure 19: The velocity vector $(x'(c), y'(c))$ (blue) decomposed into its two axis components $x'(c)$ (orange, along the x'-axis) and $y'(c)$ (green, along the y'-axis) — the two terms that appear, each multiplying a partial derivative of f, in the chain-rule sum above.

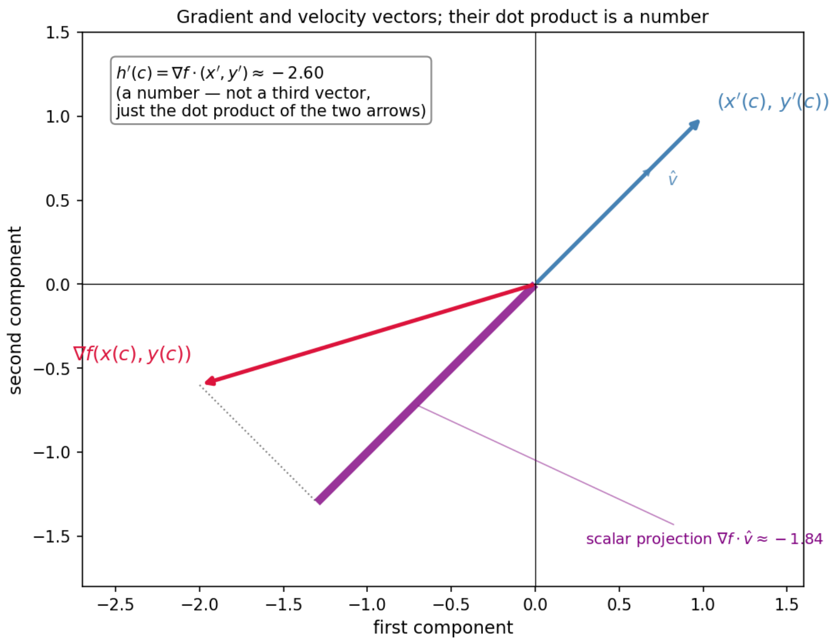

Figure 20: The two vectors of the formula shown together: the gradient $\nabla f(x(c), y(c))$ (crimson) and the velocity $(x'(c), y'(c))$ (blue), with the unit vector $\hat{v}$ along the velocity direction shown dashed. The thick purple segment is *not* a third vector: it is the scalar projection $\nabla f \cdot \hat{v}$, a length measured along the $\hat{v}$-axis (negative here, since the segment points opposite to $\hat{v}$), obtained by dropping a perpendicular (dotted grey) from the tip of ∇f onto the line through $\hat{v}$. The quantity we actually want, $h'(c) = \nabla f \cdot (x', y')$, is a different number still — the dot product with the *un-normalised* velocity vector, $|(x', y')|$ times the scalar projection shown — and is reported separately, in the text box, precisely to avoid drawing a number as though it were a vector. In general, $\nabla f \cdot v = |\nabla f|\,|v| \cos\alpha$, where α is the angle between the two vectors; when they are perpendicular, $h'(c) = 0$, matching the fact that gradients are orthogonal to level curves.

Specialising to the envelope theorem: $y(c) = c$**.** The entire series above was deliberately general, with $x(c)$ and $y(c)$ both free functions of c. The connection to §1.1–§1.2 is the special case where the *second* variable is the identity: $y(c) \equiv c$, so $y'(c) \equiv 1$ for every c. Relabel $x(c) \to x^*(c)$ (the optimiser) and write $f(x, c)$ in place of $f(x, y)$. The boxed chain-rule formula above becomes

$$\frac{d}{dc}\, f\big(x^*(c),\, c\big) = \frac{\partial f}{\partial x}\big(x^*(c), c\big) \cdot \underbrace{\frac{dx^*}{dc}}_{= x'(c)} + \frac{\partial f}{\partial c}\big(x^*(c), c\big) \cdot \underbrace{1}_{= y'(c)},$$

which is exactly the chain-rule expansion used to prove the elementary envelope theorem. The first term vanishes by the first-order condition $\partial f/\partial x\,(x^*(c), c) = 0$, leaving $V'(c) = \partial f/\partial c\,(x^*(c), c)$, the boxed result of §1.1. In the dot-product language of Figure 20: the velocity vector in this special case is $(x'(c),\, 1)$ — always a vector with second component exactly 1 — and the vanishing of the first term is the geometric statement that the *first component* of ∇f (the $\partial f/\partial x$ direction) contributes nothing to the dot product at the optimiser, because that component of ∇f is itself zero there. The full generality of the picture series was needed to see clearly that this vanishing is special to the optimiser, and not a property of the dot-product formula itself, which in general has both terms contributing.

1.4 The Probabilist's continuation: the dot product as a function of c, drawn on common abstract axes

The picture series above (§1.3) reached the dot product $\nabla f \cdot (x', y')$ by drawing the two vectors in their own native planes and resorting to a projection construction to depict the resulting number. The Probabilist's continuation, from the same morning, reaches the identical place by a more direct route: put both vectors in one common, abstract coordinate system from the start, and treat the dot product not as a single projection diagram but as an ordinary function of c.

Step 1: common abstract axes (a, b)**.** Rather than keeping the velocity vector in an (x', y')-plane and the gradient in a separately-labelled plane, relabel both as living in one shared abstract plane with axes (a, b) — emphasising that *both* vectors are just ordered pairs of real numbers, and the labels x', y' versus $\partial_x f, \partial_y f$ are bookkeeping, not a difference in the kind of object.

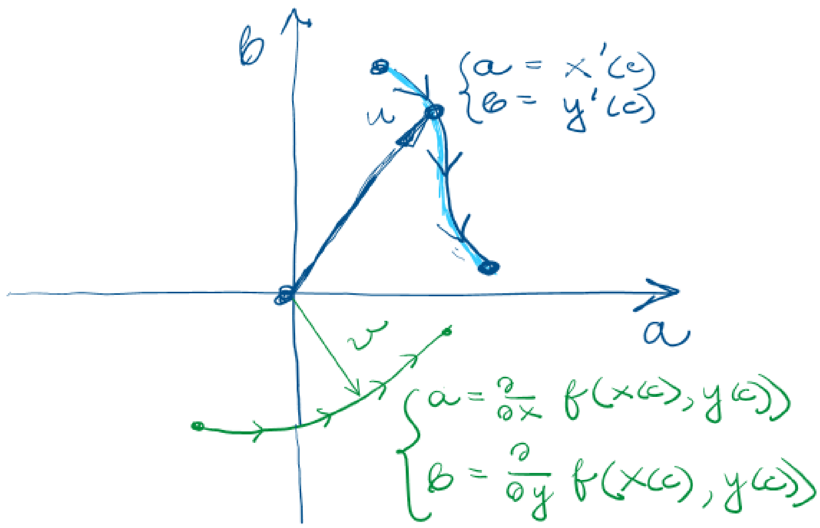

Figure 21: Both parametric curves placed in one common (a, b)-plane. Blue: $u(c) = (a, b) = (x'(c), y'(c))$, the velocity vector, traced as c varies. Green: $v(c) = (a, b) = \big(\partial_x f(x(c), y(c)), \partial_y f(x(c), y(c))\big)$, the gradient vector, also traced as c varies. Putting both in the same (a, b)-plane (rather than in two visually separate planes, as in Figures 13 and 20) makes plain that u and v are directly comparable objects of the same type, both living in $\mathbb{R}^2$.

Step 2: fix one $c = c_0$ and isolate the two vectors. Pin down a single parameter value c_0 and draw only the two resulting vectors $u(c_0)$ and $v(c_0)$, dropping the rest of each curve. Crucially, the dot product $u(c_0) \cdot v(c_0)$ is then shown not as a geometric arrow but as a labelled point on a separate number line below — directly resolving the earlier confusion (§1.3, on the corrected Figure 20) about whether the dot product should be drawn as a vector at all. It should not: it is a single real number, and a number line is the correct picture for it.

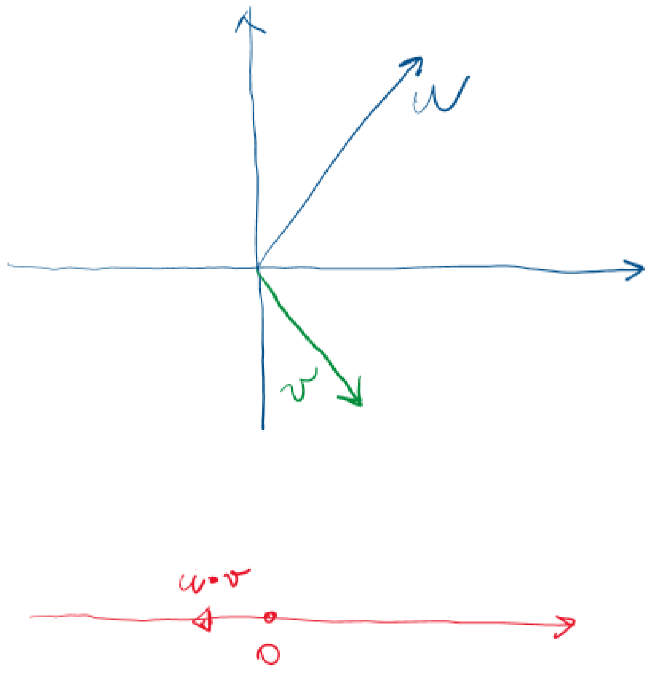

Figure 22: For one fixed $c = c_0$: the vector u (blue) and the vector v (green) in the (a, b)-plane. Below, a separate number line carries the single point $u \cdot v$ — the dot product, drawn exactly as what it is: a scalar, located on $\mathbb{R}$, not an arrow in $\mathbb{R}^2$.

Step 3: let c vary — the dot product becomes a function $\mathbb{R} \to \mathbb{R}$. The dot product itself, $(\cdot\,,\cdot) : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$, is a fixed bilinear functional — it does not depend on c. What depends on c are its two arguments, $u(c)$ and $v(c)$. Composing,

$$c \;\longmapsto\; u(c) \cdot v(c) \;=\; h'(c)$$

is an ordinary function from $\mathbb{R}$ to $\mathbb{R}$, exactly like any other univariate function — and so it can be drawn as an ordinary 2-d graph over the c-axis, with no vectors or planes needed anymore.

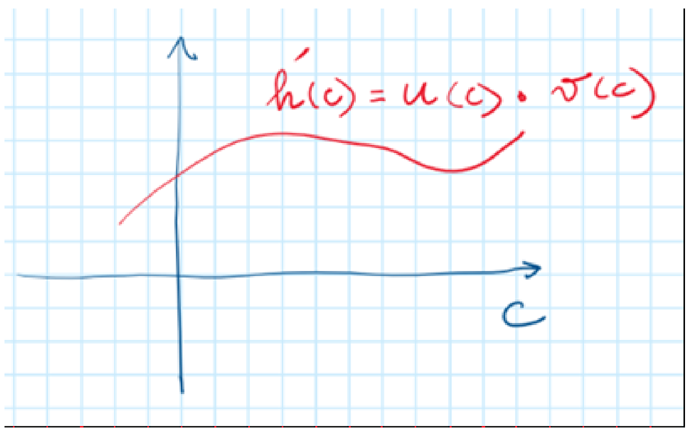

Figure 23: The function $c \mapsto h'(c) = u(c) \cdot v(c)$, plotted directly as a curve over the c-axis. All of the vector geometry of Figures 21–22 has been absorbed into evaluating the dot product at each c; the result is just an ordinary real-valued function of one real variable, no different in kind from $x(c)$ or $y(c)$ themselves.

Why this route is cleaner. The projection construction of §1.3 (Figure 20) is geometrically informative — it shows *why* the dot product has the value it does, via $|\nabla f|\,|v| \cos\alpha$ — but it requires care to avoid mis-drawing the resulting scalar as a vector, exactly the error flagged earlier in this section. The Probabilist's route sidesteps that risk entirely: by drawing $u \cdot v$ on a number line at a single c (Figure 22), and then immediately treating $h'(c)$ as an ordinary function of c (Figure 23), there is no step at which a scalar quantity is ever rendered with an arrowhead. The two approaches are complementary: the projection picture explains the *value* of the dot product geometrically; this picture keeps the *type* of the dot product (a number, varying with c) unambiguous throughout.

1.5 Three problems and the triangle connecting them

We state three problems separately, in their own standard notation, and then connect them in pairs.

Problem 1: constrained optimisation, parametrised by a right-hand side c. Maximise $f(x)$ subject to $g(x) = c$, where c is a parameter of the constraint's right-hand side. The Lagrangian is

$$\mathcal{L}(x, \lambda; c) = f(x) - \lambda\big(g(x) - c\big).$$

For each c, a stationary point $\big(x^*(c), \lambda^*(c)\big)$ solves $\nabla_{x,\lambda}\mathcal{L} = 0$. Define the *value function*

$$V(c) := \mathcal{L}\big(x^*(c), \lambda^*(c);\, c\big) = f\big(x^*(c)\big),$$

a function of c alone. It can be differentiated in c; we will need $V'(c)$ below.

Remark 3. This is the single-variable Lagrangian, matching Part 5b-a throughout. It is deliberately *not* the two-variable $f(x, y)$ of the chain-rule picture series (§1.3), where a genuine second variable y appears alongside a path $x(c), y(c)$. Side A below needs only one variable, so we state Problem 1 with one variable from the start.

Problem 2: the envelope theorem. In the simplest setting — $\max_x f(x, c)$, unconstrained, no g — write $x^*(c) = \arg\max_x f(x, c)$ and $V(c) = f(x^*(c), c)$. The envelope theorem states

$$V'(c) = \frac{\partial f}{\partial c}\Big(x^*(c), c\Big),$$

proved by the chain rule: the term $\partial f/\partial x \cdot dx^*/dc$ vanishes because $\partial f/\partial x = 0$ at $x^*(c)$. The same cancellation, applied jointly in (x, y, λ), gives Problem 1's value function

$$V'(c) = \frac{\partial \mathcal{L}}{\partial c}\bigg|_{x^*(c), y^*(c), \lambda^*(c)} = \lambda^*(c),$$

since $\partial\mathcal{L}/\partial c = \lambda$. This is the *shadow price* reading: λ^* is the rate of change of the optimised value as the constraint level moves.

Problem 3: the score test, informally. Take a log-likelihood $\ell(\theta)$ and a simple null $H_0 : \theta = \theta_0$. The score is $s(\theta) = \ell'(\theta)$. Under H_0, the identity $\mathbb{E}_{\theta_0}[s(\theta_0, X)] = 0$ holds, and by the CLT the sample average score concentrates around zero and shrinks at rate $1/\sqrt{n}$ as the sample size grows. The test rejects H_0 when the observed score $s(\theta_0)$ is too far from zero relative to this shrinking spread — formalised as $\xi_{LM} = s(\theta_0)^2/\mathcal{I}(\theta_0) \xrightarrow{d} \chi^2_1$.

Side A: Problem 1 ↔ Problem 2. Apply Problem 1 with $x \to \theta$, $f(x) \to \ell(\theta)$, and the constraint $g(\theta) = \theta = c$ (so g is the identity). The genuine difficulty: the feasible set $\{\theta = c\}$ is one point, so $\theta^*(c) = c$ trivially, with no first-order condition in θ alone — the naive match $x \to \theta$ fails on its own, since there is nothing to differentiate. The fix is to apply Problem 2's theorem not to θ alone but to the pair (θ, λ) jointly, via $\mathcal{L}(\theta, \lambda; c) = \ell(\theta) - \lambda(\theta - c)$: both first-order conditions now hold genuinely ($\ell'(\theta) - \lambda = 0$ recovers λ; $-(\theta - c) = 0$ recovers $\theta = c$), even though (θ^*, λ^*) is a saddle, not a maximum — legitimate because the chain-rule proof of Problem 2 only used the first-order condition, never concavity. Carrying this out: $V(c) = \ell(\theta^*(c)) - \lambda^*(c) \cdot 0 = \ell(c)$ directly, so $V'(c) = \ell'(c)$; and via the envelope theorem $V'(c) = \lambda^*(c)$. Equating:

$$\boxed{\lambda^*(c) = \ell'(c).}$$

The Probabilist's strengthening: the Lagrangian equals V identically, not just up to a constant. The Probabilist's notes of June 21, 2026 revisit Problem 2 once more, this time following Wikipedia's statement of the constrained envelope theorem. The upgrade relative to the further generalisation of §1.2 (which let the constraint be $g(\theta, c) = 0$ but kept $f(\theta)$ free of c) is to let f itself depend on c as well, matching Problem 2's $f(x, c)$:

$$\max_x f(x, c) \quad \text{subject to} \quad g(x, c) = 0, \qquad \mathcal{L}(x, \lambda; c) = f(x, c) - \lambda\, g(x, c).$$

For each fixed c, the Probabilist takes $\big(x^*(c), \lambda^*(c)\big)$ to be the stationary point of $\mathcal{L}(x, \lambda)$ — a function of the *two* variables x, λ at that fixed c, exactly the saddle-point picture recalled at the start of this notebook entry — and sets $V(c) := f(x^*(c), c) = \max_{g=0} f(x, c)$, as before. The envelope theorem as already stated gives the derivative identity

$$V'(c) = \frac{\partial \mathcal{L}}{\partial c}\Big(x^*(c), \lambda^*(c), c\Big). \tag{1}$$

The Probabilist's observation is that, because the constraint is an *equality* ($g = 0$, not $g \leq 0$), a strictly stronger statement holds: not merely that the two sides of (1) share a derivative, but that the underlying functions of c coincide *exactly*, with no leftover constant of integration:

$$\mathcal{L}\big(x^*(c), \lambda^*(c), c\big) = V(c) \qquad \text{for every } c. \tag{2}$$

The proof is immediate from the definition of $\mathcal{L}$ together with feasibility of the optimiser:

$$\mathcal{L}\big(x^*(c), \lambda^*(c), c\big) = f\big(x^*(c), c\big) - \lambda^*(c)\, g\big(x^*(c), c\big) = V(c) - \lambda^*(c) \cdot 0 = V(c),$$

the last step using $g(x^*(c), c) = 0$: the optimiser is feasible at *every* c, so the constraint term vanishes identically, not just at one value of c.

Differentiating (2) directly recovers (1), so (2) $\Rightarrow$ (1); the converse need not hold, since (1) alone only says the two sides agree up to an additive constant in c, while (2) pins that constant at exactly zero. As the Probabilist puts it: the two functions are not merely shifted copies of one another — they are equal, the shift being zero.

This is not new content so much as an unarticulated special case of what Side A already computed: the step "$V(c) = \ell(\theta^*(c)) - \lambda^*(c) \cdot 0 = \ell(c)$" above *is* identity (2), for the particular instance where $f(\theta)$ does not depend on c and $g(\theta, c) = \theta - c$. The Probabilist's note packages this as a general fact about equality-constrained Lagrangians, rather than a coincidence of that one computation.

Caveat (the Probabilist). Identity (2) rests on $g(x^*(c), c) = 0$ holding exactly — genuine equality. If the constraint is instead an inequality, $g(x, c) \leq 0$, only the derivative identity (1) is asserted in general; (2) may fail, even though complementary slackness still gives $\lambda^*(c)\, g(x^*(c), c) = 0$ pointwise at each c. The Probabilist flags this contrast without resolving it further here.

Side B: Problem 1 $\leftrightarrow$ Problem 3. Renaming $c \to \theta_0$, Side A's boxed identity reads $\lambda^* = \ell'(\theta_0) = s(\theta_0)$: the Lagrange multiplier at the constrained optimum *is* the score at the null. This is exactly the quantity Problem 3 squares and divides by Fisher information to build ξ_{LM}. So Problem 1's λ^* is not merely analogous to Problem 3's score — it is the same number, under two names, which is why the test is called both the "score test" and the "LM test."

Side C: Problem 2 $\leftrightarrow$ Problem 3. The shadow-price reading of Side A ($\lambda^* = V'(\theta_0)$, the sensitivity of the optimised log-likelihood to where we place the null) gives the score a second interpretation, beyond "slope of ℓ at θ_0": it is *how much the constrained optimum would improve if the null were nudged*. A score near zero means the optimised value is insensitive to nudging θ_0 — the null is already where the unconstrained optimum would be, consistent with H_0. A large score means nudging θ_0 would improve the fit substantially — the data pull away from the null, exactly what Problem 3's test is built to detect. The CLT shrinkage in Problem 3 ($s(\theta_0) \to 0$ at rate $1/\sqrt{n}$ under H_0) is then read, via Side A, as: the shadow price of the constraint vanishes asymptotically when the constraint is true.

What does not change. Problem 3's test statistic needs none of Problems 1–2 to be computed — as emphasised in Part 5b-a, §2.7. The triangle above is explanatory: it shows the same number, $\lambda^* = s(\theta_0)$, arising from three different questions, and gives the score test's central quantity a economic name (shadow price) that the bare formula does not supply on its own.

1.6 A historical note: which came first, score or multiplier?

The triangle above raises a natural question: is the Lagrangian apparatus actually needed to define or compute the score test, or is it decoration? The answer, confirmed by the history, is that it is decoration — but the chronology is the reverse of what one might guess.

Rao (1948) came first, with no Lagrangian at all. C. R. Rao's 1948 paper introduces the score test directly: the statistic is built from the score $s(\theta_0)$ and the Fisher information $\mathcal{I}(\theta_0)$, with $\xi = s(\theta_0)^2/\mathcal{I}(\theta_0) \xrightarrow{d} \chi^2_1$ under H_0. No constrained optimisation, no Lagrangian, no multiplier appears in this original formulation — it is Problem 3 above, stated and proved on its own terms.

Silvey (1959) came eleven years later, adding the Lagrangian framing. Aitchison and Silvey (1958) and Silvey (1959) approached the same problem from the constrained-MLE side: maximise the log-likelihood subject to the null's restrictions, form the Lagrangian, and observe that the resulting multiplier λ^* is (asymptotically) distributed the same way as Rao's score statistic. Silvey's 1959 paper is the source of the name "Lagrange Multiplier test." So the order is: *score test first, Lagrangian re-derivation second* — not the reverse.

Correcting one detail, confirming the main point. This means the natural guess — that the test was first defined via λ^* and only later "unburdened" into the score form — has the chronology backwards: Rao's original score form never carried Lagrangian baggage to begin with; Silvey's Lagrangian version was an *additional*, later route to the same number, not a simplification of an earlier, more complicated one. But the substantive conclusion is exactly right: computing and justifying the test requires only $s(\theta_0)$ and $\mathcal{I}(\theta_0)$, both of which are obtained by differentiation alone, with no optimisation or stationarity condition involved (Problem 3, on its own, needs none of Problem 1's machinery). The Lagrangian route (Side A above) shows that λ^* — obtained by solving a constrained-optimisation stationarity system — happens to equal $s(\theta_0)$, but as established earlier in this document, that stationarity system is solving a problem whose feasible set is a single point:

there is no genuine optimisation content in it, and the "answer" it returns is no different from what direct differentiation already supplies.

Why both names survived, rather than one replacing the other. There is no documented single event where the field renamed the test from "LM" to "score"; both names remain in active use today, split roughly along disciplinary lines — the score-test name is more common in statistics (following Rao), the LM-test name more common in econometrics (following Silvey, and popularised further by Breusch and Pagan's 1980 use of it for the heteroscedasticity test that motivated this entire document). The coexistence reflects two independent discoveries of the same asymptotic statistic, each retaining the name attached to its own derivation route, rather than a deliberate later simplification of one name into the other.